

A THOUSAND BRAINS

A New Theory of Intelligence



Based on the work of Jeff Hawkins | Foreword by Richard Dawkins

The brain sits in darkness... but constructs the light.

The Only Matter That Knows

The universe is vast, old,
and mostly silent.

If a tree falls and no one
hears it, it creates air
vibrations, not sound.

Sound is a creation of
the brain.

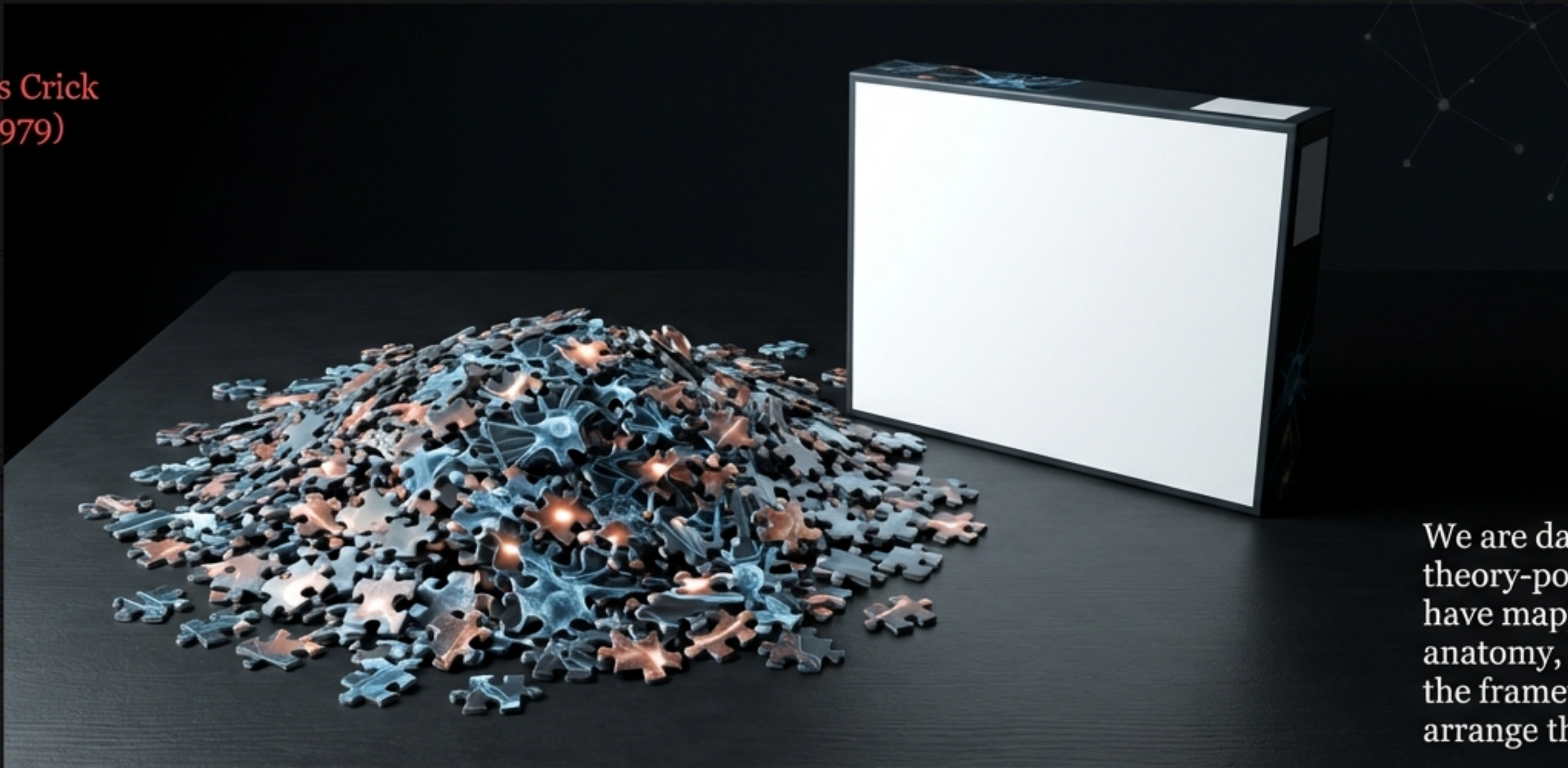


A 3-pound mass of cells
is the only thing in the
universe that **knows** the
universe exists.

Without it, the cosmos
would come into and out
of existence without a
witness.

The Puzzle vs. The Picture

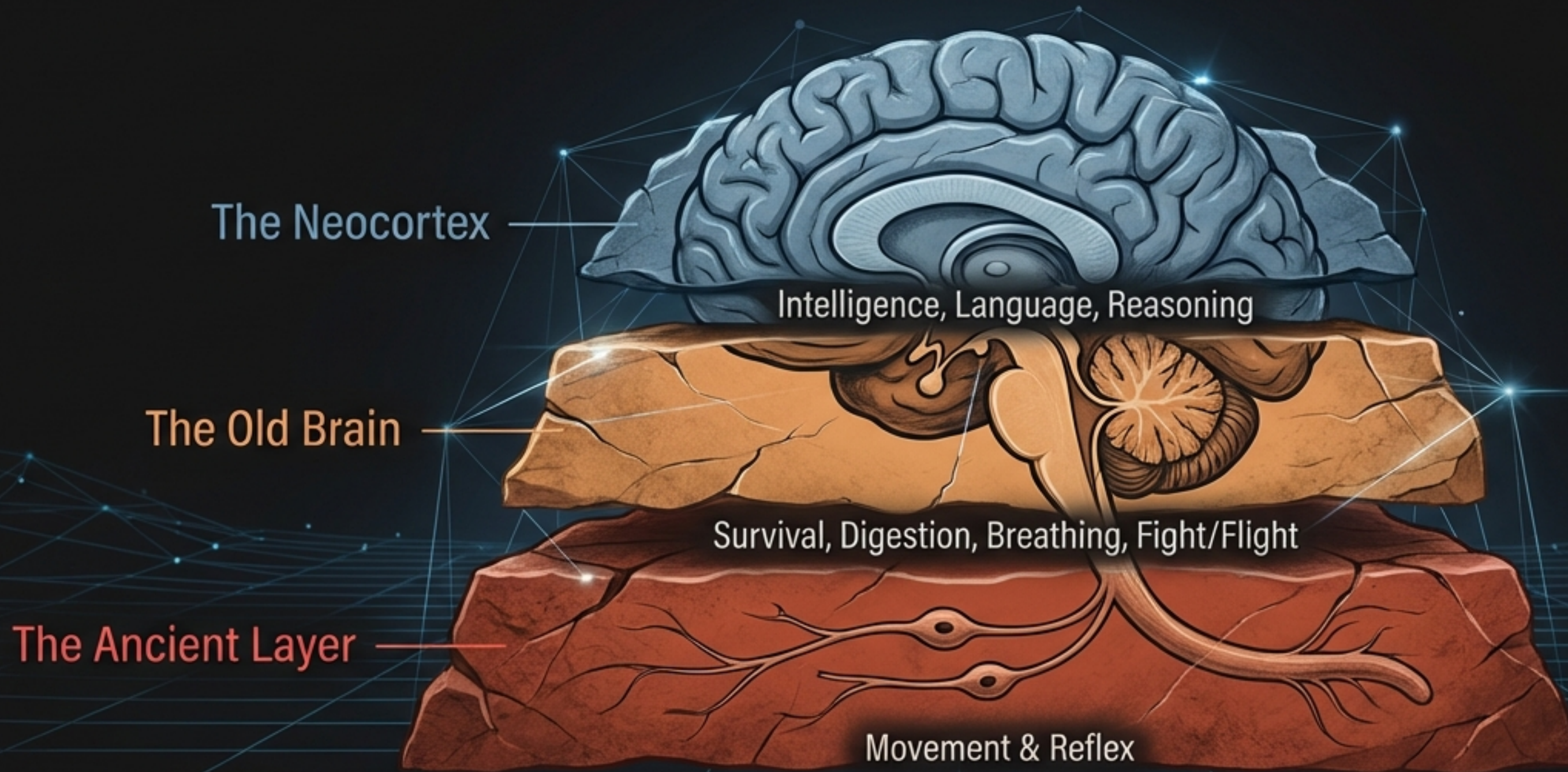
The Francis Crick
Problem (1979)



We are data-rich but theory-poor. We have mapped the anatomy, but we lack the framework to arrange the pieces.

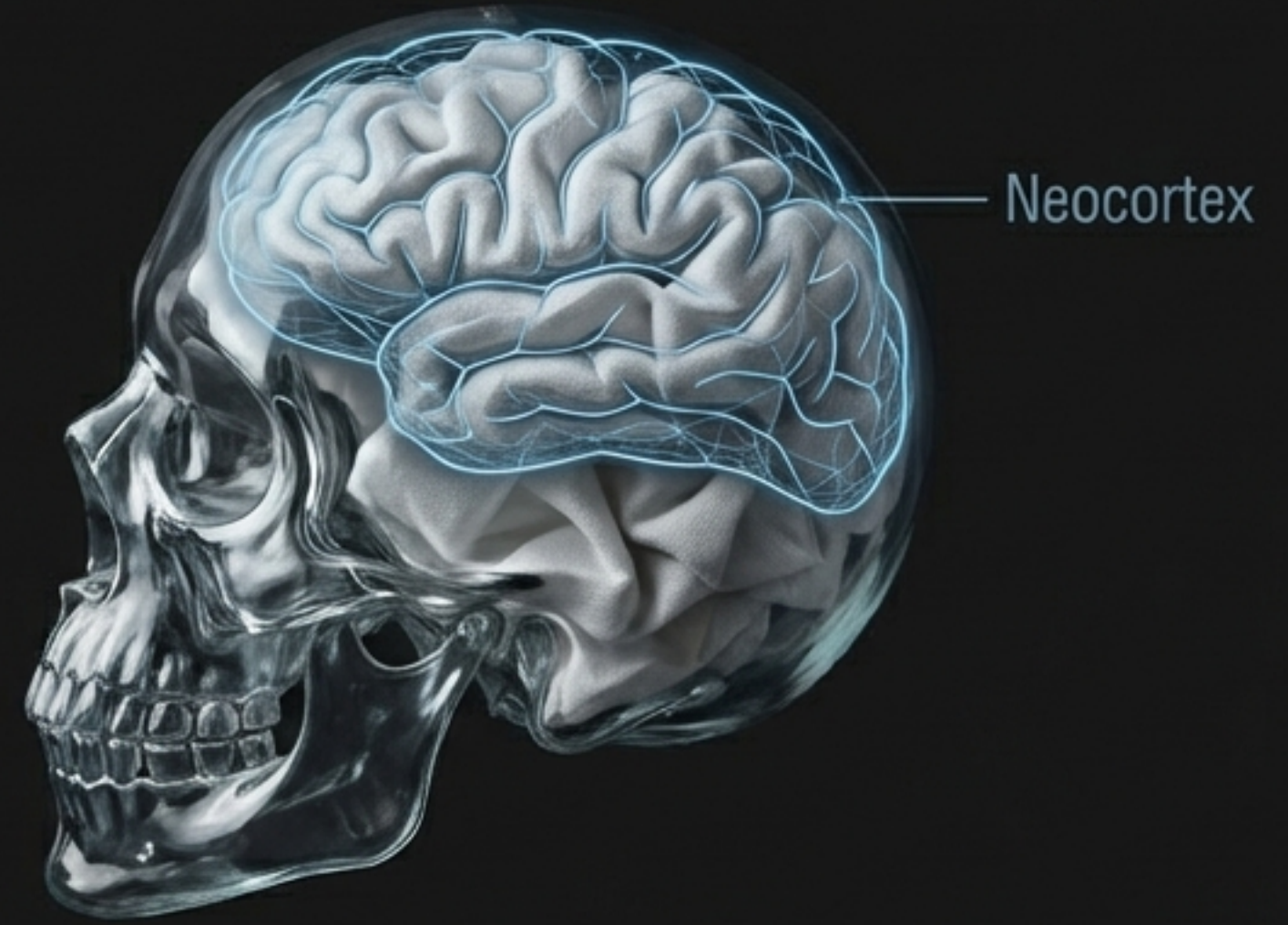
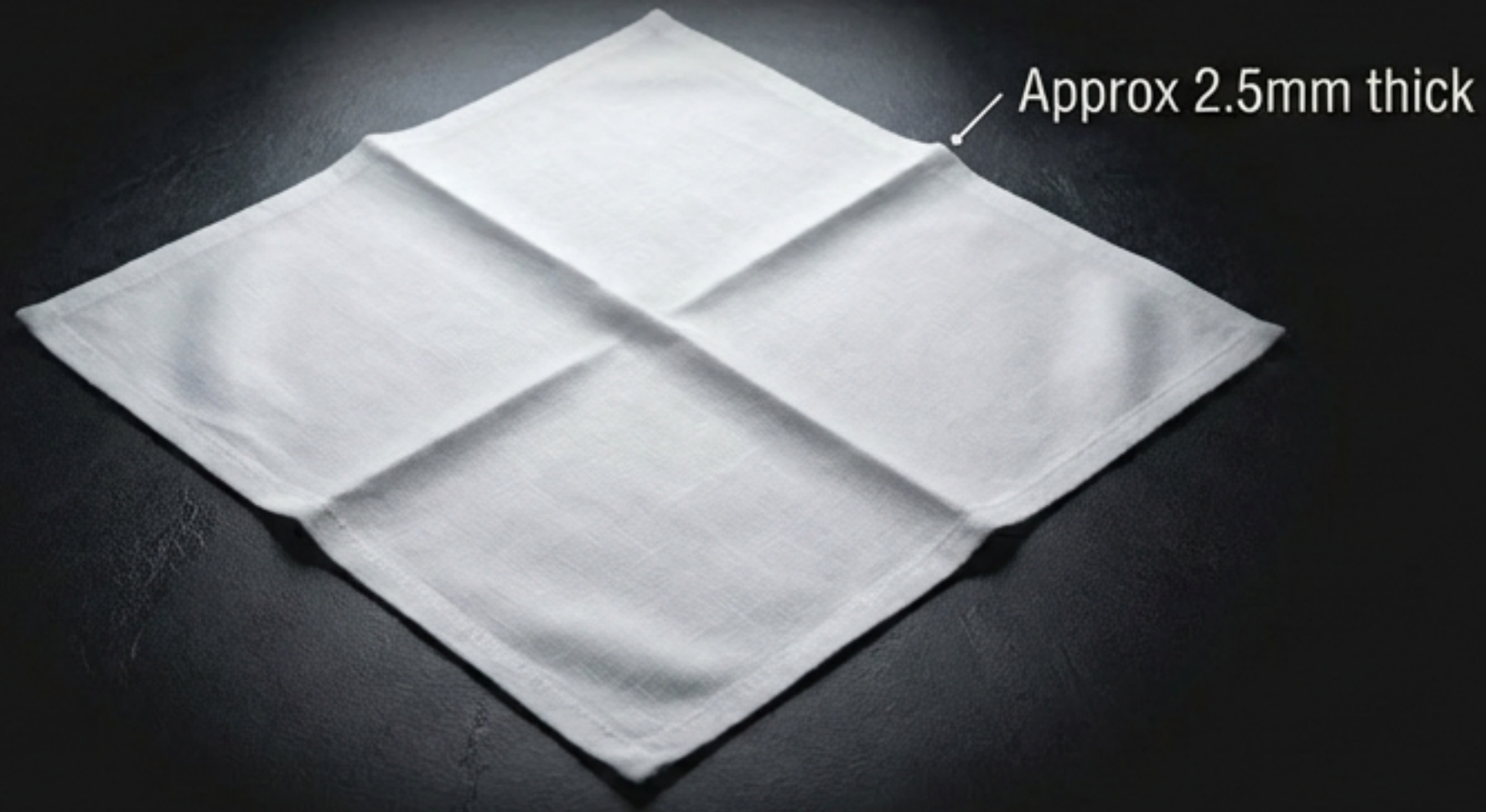
“In spite of the steady accumulation of detailed knowledge, how the human brain works is still profoundly mysterious.” — Francis Crick

Evolution by Addition, Not Replacement



Evolution never redesigns from scratch. It builds new parts on top of old parts.

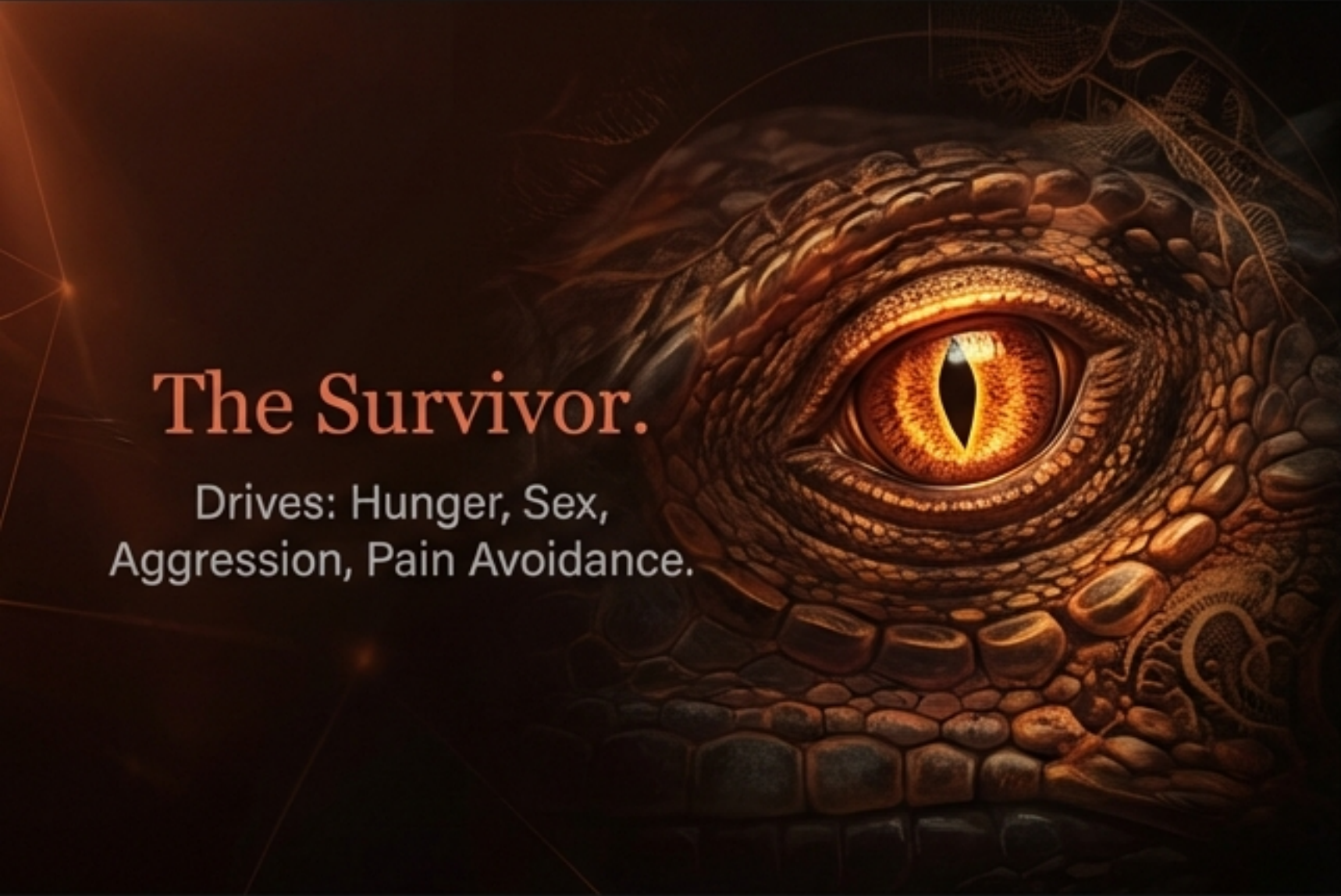
The Seat of Intelligence: The Neocortex



Volume: 70% of the brain | Thickness: 2.5mm.

If you ironed out the folds of your neocortex, it would be the size of this dinner napkin.
That sheet of cells is reading this slide right now.

The Roommates



The Survivor.

Drives: Hunger, Sex,
Aggression, Pain Avoidance.

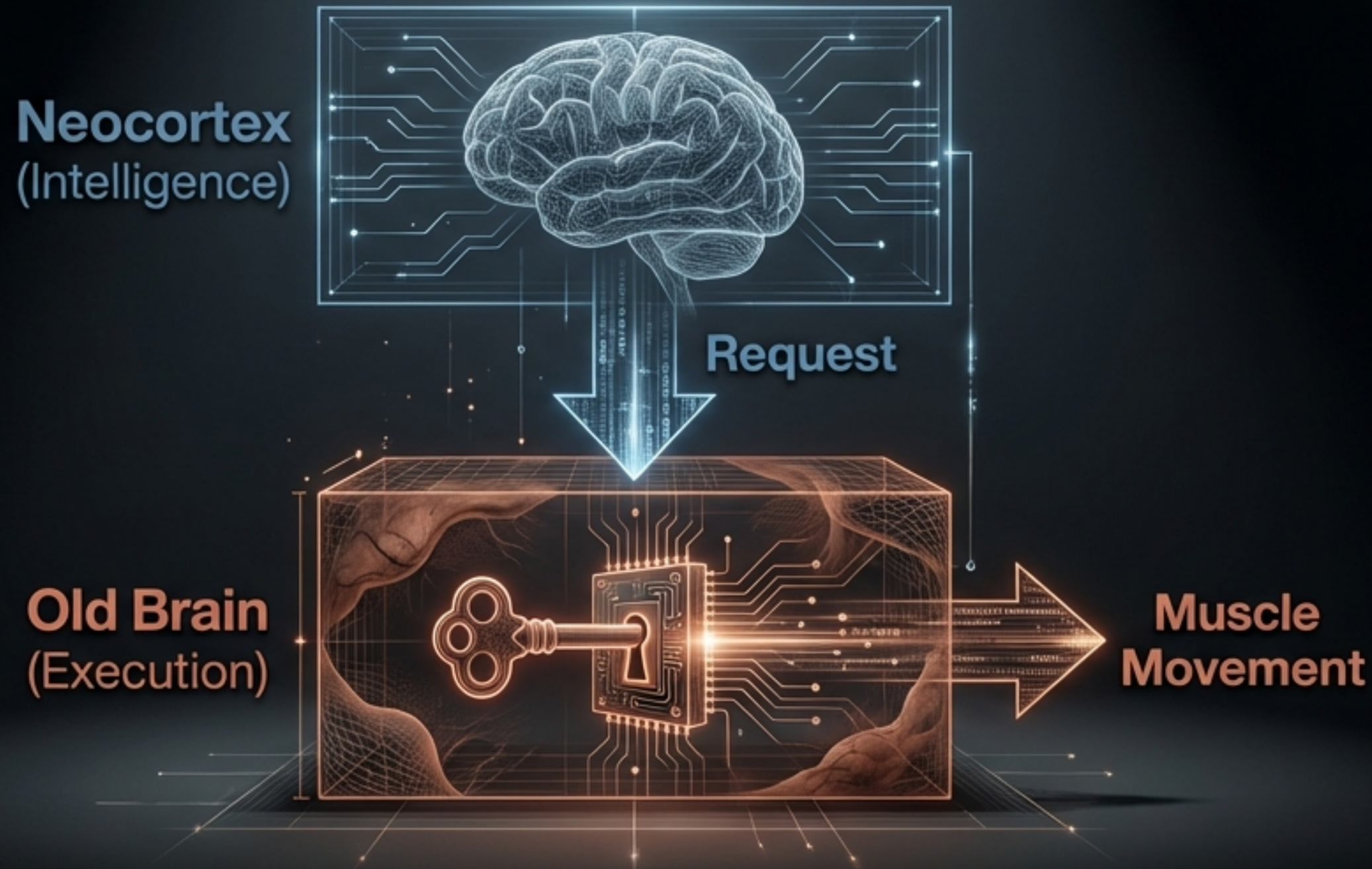


The Planner.

Drives: Logic, Future
Goals, Social Restraint.

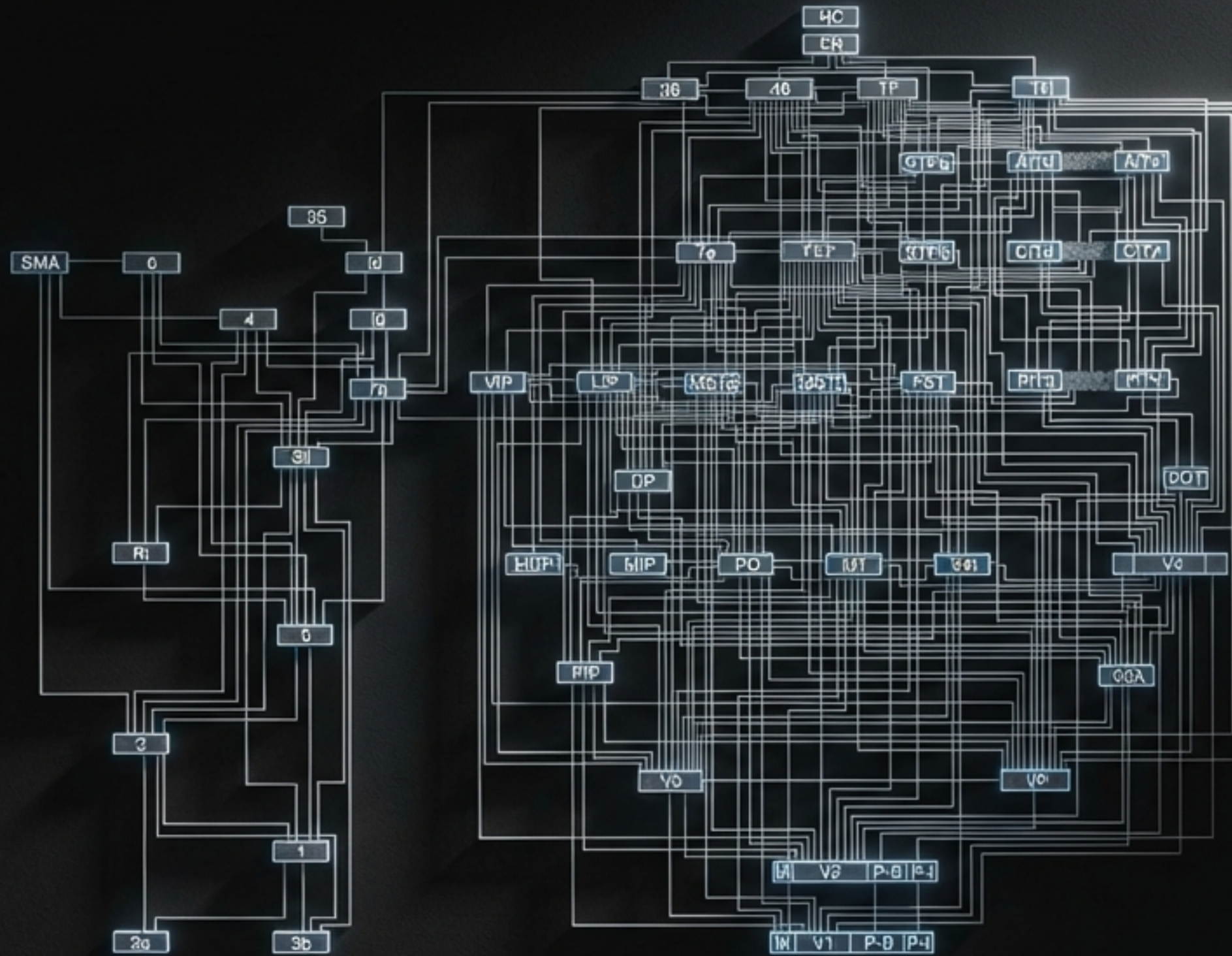
They live in the same skull but have different agendas.
The Old Brain controls the emotions; the New Brain tries to guide it.

The Illusion of Control



Fact: No cells in the **neocortex** connect directly to muscles. The intelligent mind must **request** action from the **primitive brain**. If the **Old Brain** disagrees (e.g., holding breath until oxygen drops), it **takes the keys back**.

The Map of Complexity



For decades, this map represented our understanding: a **chaotic hierarchy** of specialized regions.

Visual info here, **touch** info there.

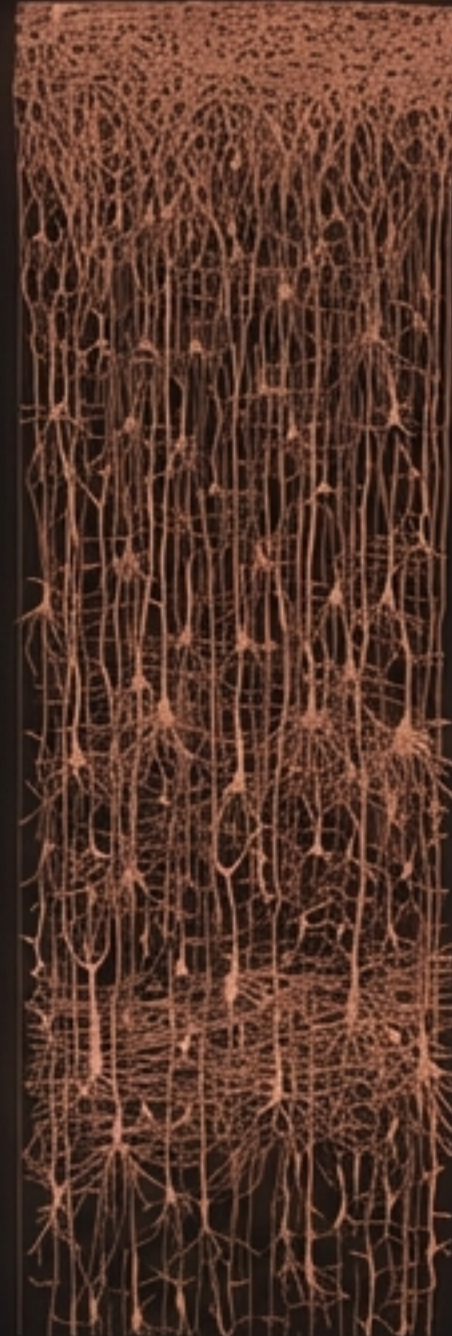
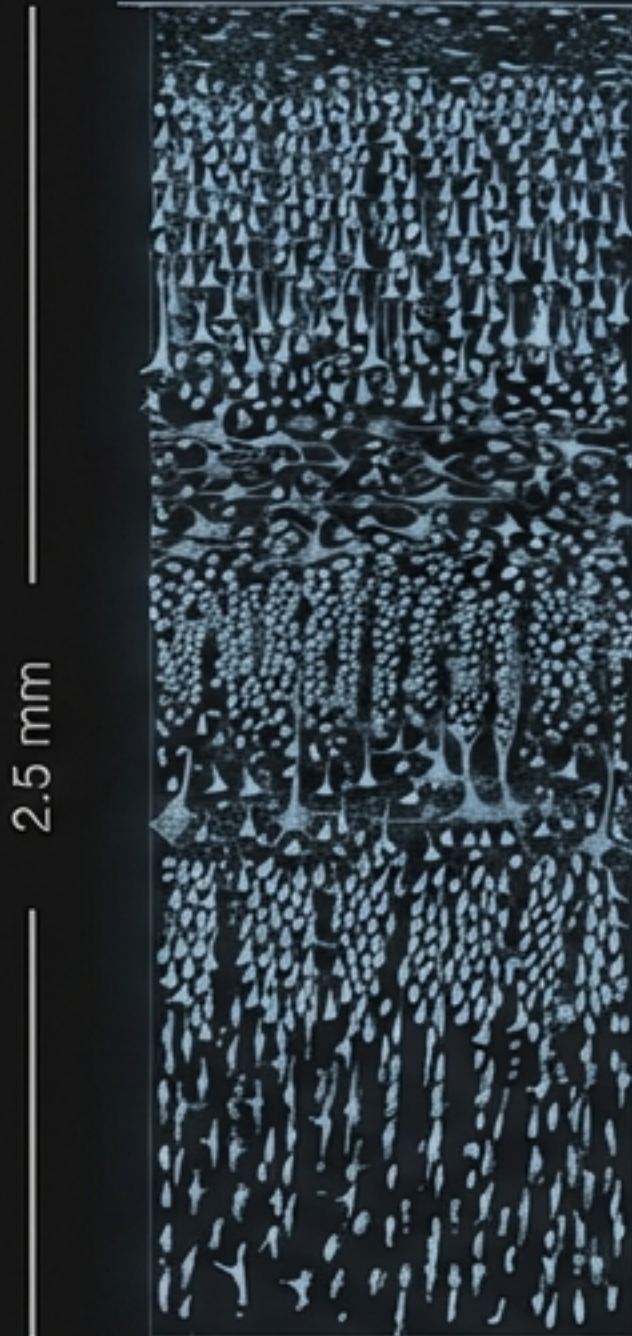
It looks less like a designed machine and more like a **plate of spaghetti**.

The Felleman & Van Essen Wiring Diagram (1991)

The Uniformity of Gray Matter

Vision Cortex

Language Cortex



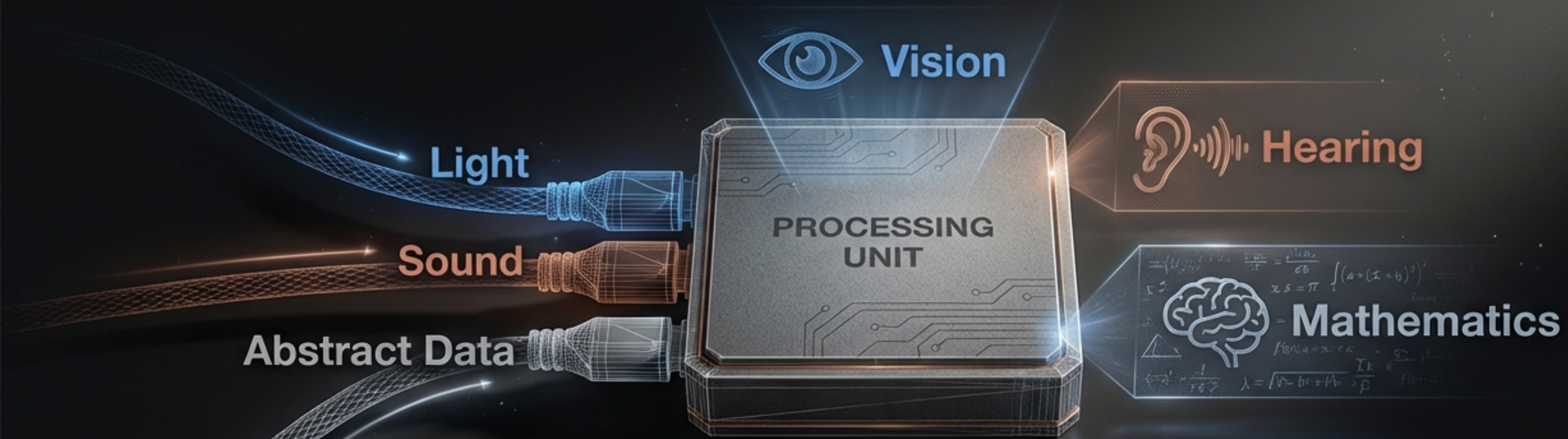
The Mystery:

Under a microscope, a slice of the brain responsible for vision looks identical to a slice responsible for language, math, or touch.

If the functions are so different, why is the hardware identical?

Cajal, 1899

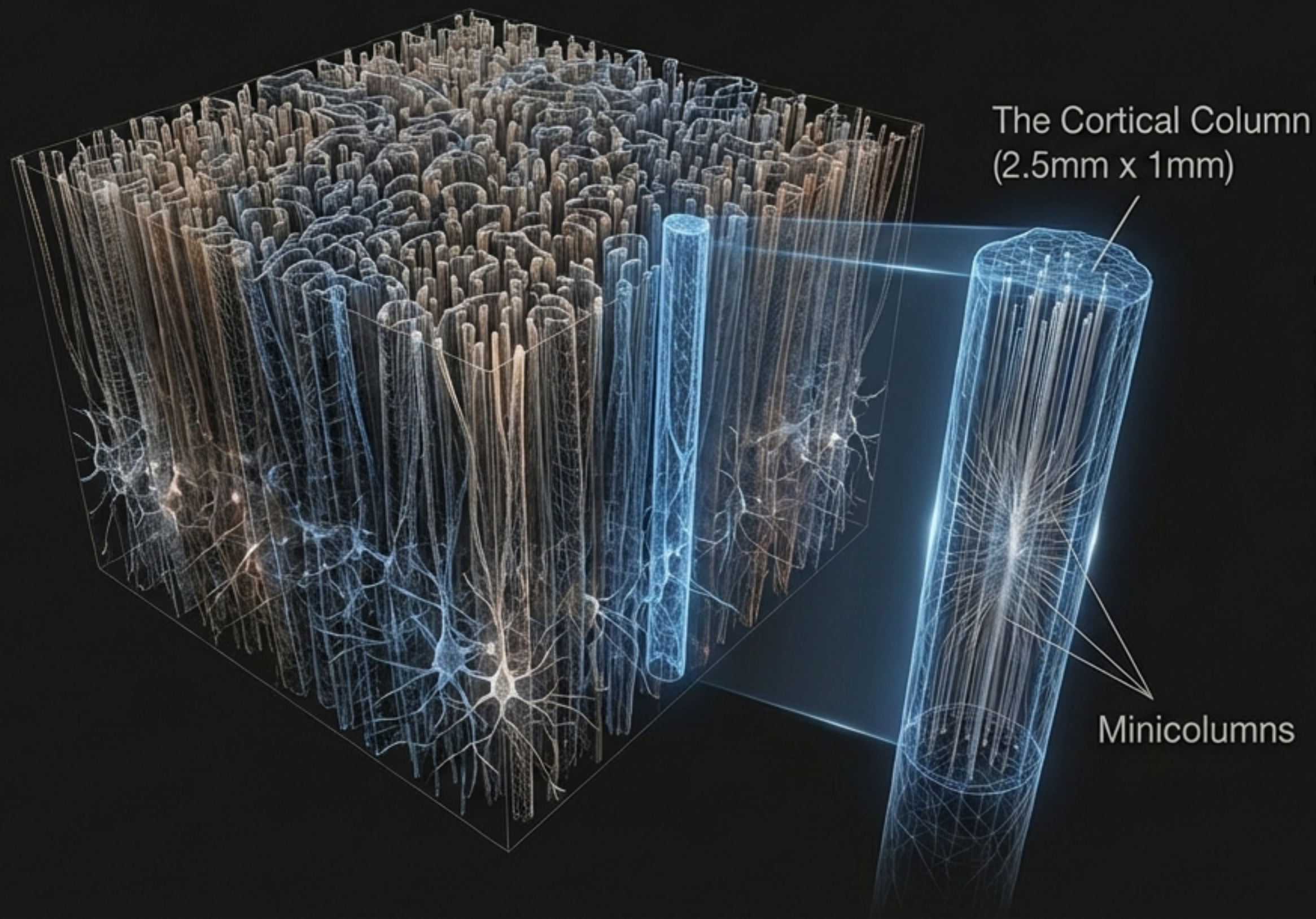
Vernon Mountcastle's Big Idea



“There is nothing intrinsically motor about the motor cortex, nor sensory about the sensory cortex.” — Vernon Mountcastle.

The brain uses ONE universal algorithm for everything. Intelligence is not a set of tools; it is a flexible learning method.

The Unit of Intelligence: The Cortical Column

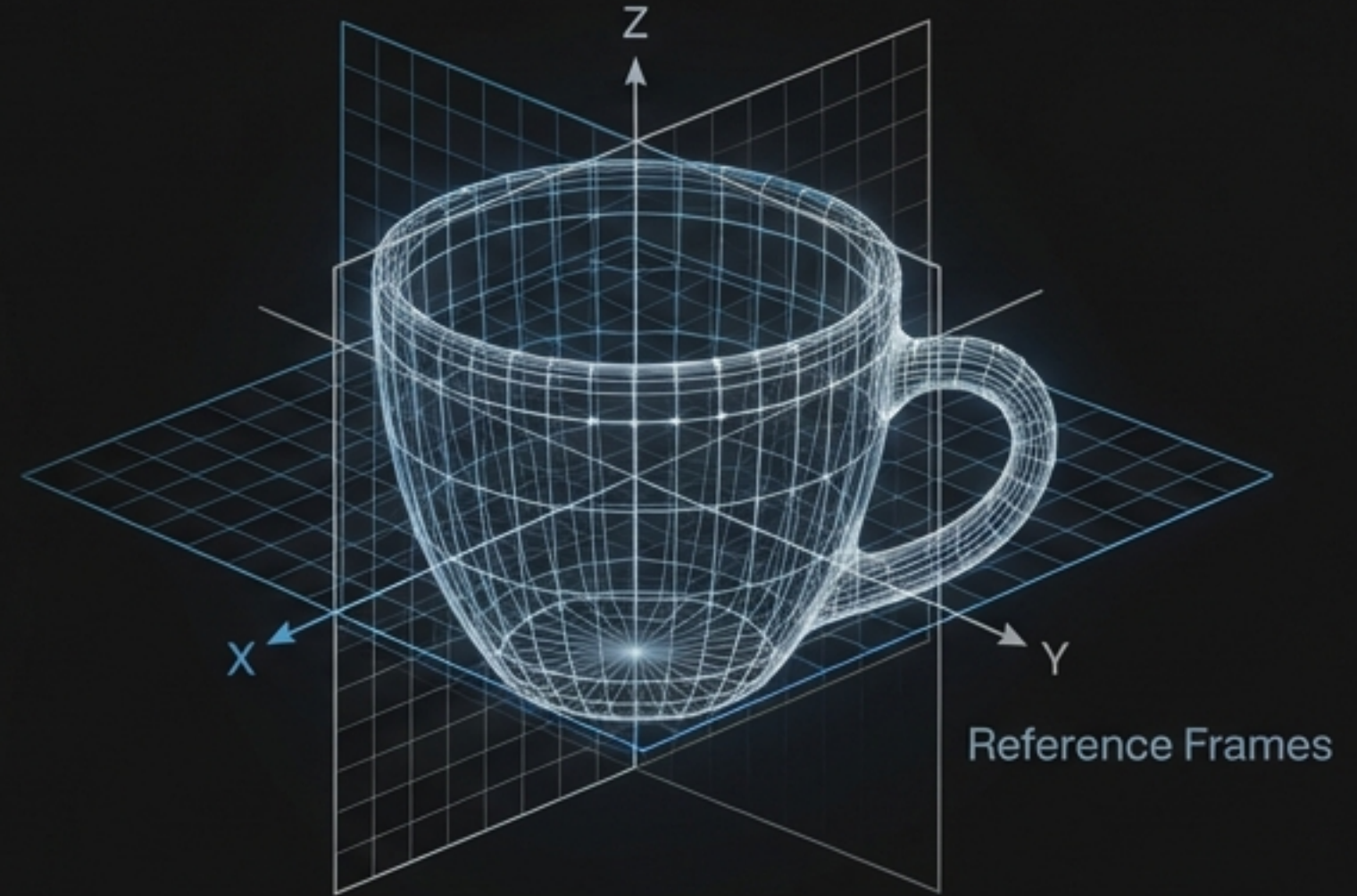


The Cortical Column
(2.5mm x 1mm)

The brain grew large not
by inventing new parts,
but by copying this single
computing unit
150,000 times.

Minicolumns

Reference Frames: The Grid Lines of the Mind



To know a city, you use a map with grid lines (Latitude/Longitude).
To know an object, your brain builds a 3D structure using Reference Frames.

Thinking is a form of movement. We navigate through abstract concepts the same way we navigate physical space.

The Thousand Brains Theory in Steel Blue



Distributed Consensus: There is no central processor.
Thousands of **cortical columns** generate their own models of the world.
What we perceive is the **consensus vote** of these thousands of brains.

Why There is No 'I' in AI (Yet)



Pattern Matching.
Good at static games.



World Model.
Understands cause, effect, and physics.

Current **AI** maps inputs to outputs.

True Intelligence understands **the structure of the world**.

To get to **AGI**, we don't need more data; we need machines that use **Reference Frames**.

The Myth of the Evil Machine



Fear, greed, and dominance are biological drives from the Old Brain, evolved for gene replication. A machine built on Neocortical principles (Pure Intelligence) has no Old Brain. It has no drive to survive or subjugate unless we program it to.

The Real Existential Risk



The Old Brain + The New Brain Weapons.

We are god-like technology wielded by primitive institutions. The danger isn't that machines will become like us; it's that we remain like us while wielding the power of machines.

The Choice: Genes vs. Knowledge



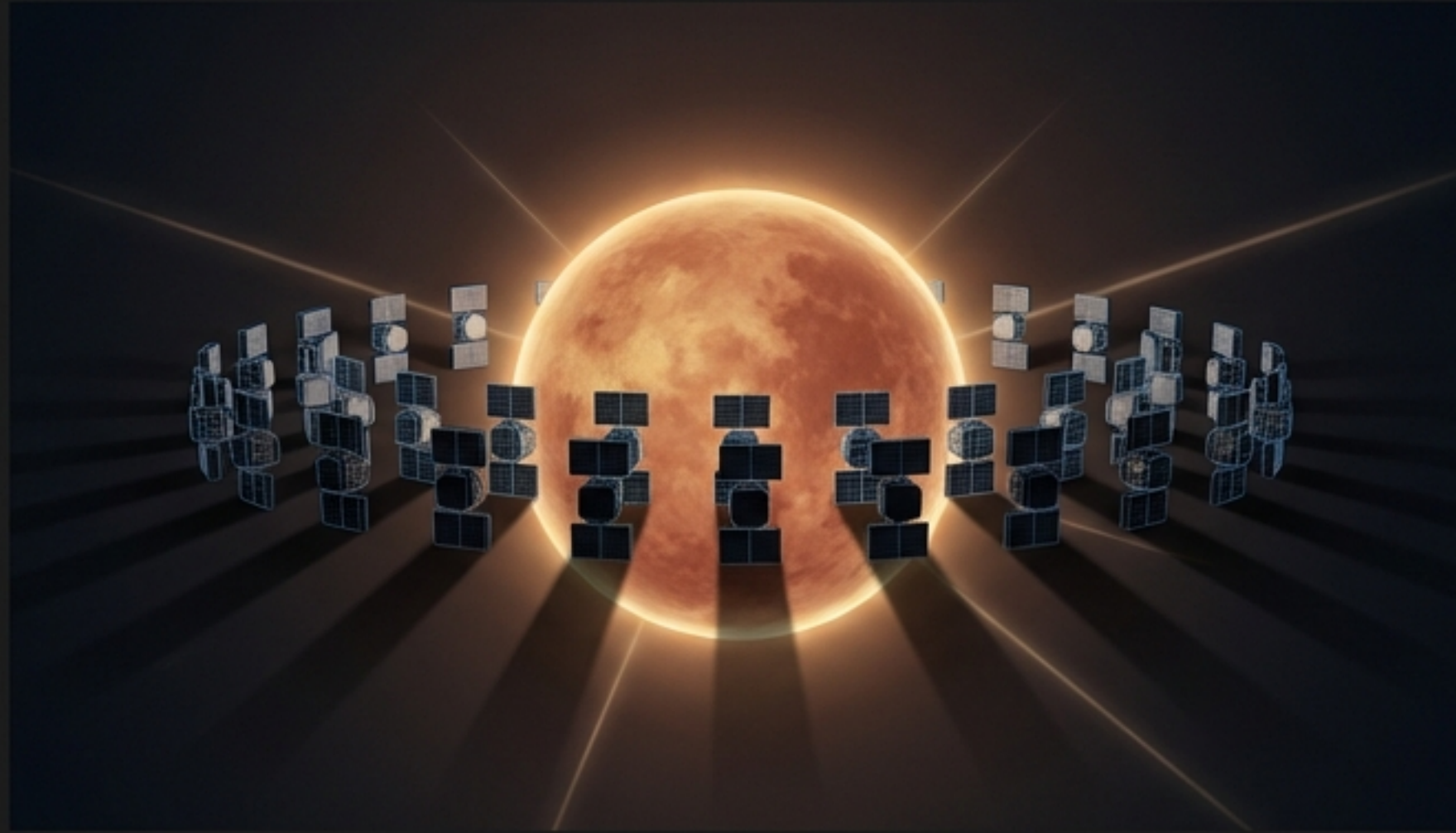
Serve the
Replicators
(Survival)



Serve the
Universe
(Understanding)

We are the first species with a choice.
Do we live to pass on our genes?
Or do we live to understand the universe?

The Cosmic Tombstone (Helvetica Now Display)



The Sun Blocker Signal

Civilizations are fireflies in the night. To leave a legacy that outlasts our biology, we could arrange satellites to dim the sun in a mathematical series (like prime numbers). A signal visible for light-years, lasting for billions of years.

Final Thoughts



“The brain sits in darkness, apprehending the outside world only through a hailstorm of nerve impulses... The Universe can live and work and plan, at last made God within the mind of man.” — Adapted from Richard Dawkins / Julian Huxley

Read “A Thousand Brains” by Jeff Hawkins.
Join the shift from biological evolution to intelligent evolution.